CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – : Big Data Analytics

Practical File – ODD Semester 2026-27

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Practical 9

1. Problem Definition

TrendPulse Social Media Analytics tracks social media activity from platforms including Twitter/X, Instagram, Facebook, and LinkedIn. Large numbers of posts contain hashtags related to technology, sports, entertainment and other current topics.

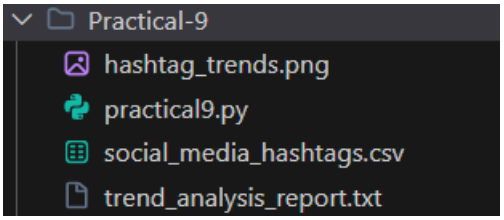
The objective of this practical is to apply PySpark for processing hashtag data and to determine frequently discussed and emerging topics.

The analysis includes hashtag grouping, occurrence counting, popularity ranking, daily and regional trend checks, category analysis and visualization.

Note: The dataset used here is a generated/sample dataset prepared for educational analysis.

PART A — Practical Execution

Step 1: Creating Folder and files



Step 2: Creating Spark Session

```
============================================================
 TASK 1: CREATING SPARK SESSION
============================================================
WARNING: Using incubator modules: jdk.incubator.vector
26/09/27 17:08:49 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see
 https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/09/27 17:08:50 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark Version   : 4.2.0
Spark App Name  : Social-Media-Trend-Detection
Master URL      : local[*]
```

Step 3: Loading Social Media Hashtag Dataset

```
============================================================
 TASK 2: LOADING SOCIAL MEDIA HASHTAG DATASET
============================================================
Dataset Path    : C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-9\social_media_hashtags.csv
Dataset loaded successfully.
Total Number of Records: 650

First 10 Records:
+----------+---------+--------------+-------------+
|Timestamp |Platform |Hashtag       |Region       |
+----------+---------+--------------+-------------+
|2026-03-04|LinkedIn |#Netflix      |Latin America|
|2026-03-04|Facebook |#CloudComputing|North America|
|2026-03-01|Instagram|#Netflix      |Asia-Pacific |
|2026-03-04|Facebook |#CloudComputing|Middle East  |
|2026-03-01|LinkedIn |#Tech         |Middle East  |
|2026-03-01|LinkedIn |#Movies       |Europe       |
|2026-03-01|LinkedIn |#Hollywood    |Europe       |
|2026-03-04|LinkedIn |#Netflix      |North America|
|2026-03-04|LinkedIn |#AI           |North America|
|2026-03-05|Twitter/X|#AI           |North America|
+----------+---------+--------------+-------------+
only showing top 10 rows
```

Step 4: Displaying Hashtag Records

```
============================================================
 TASK 3: DISPLAYING HASHTAG RECORDS
============================================================
Sample Hashtag Records (First 10 rows):
+----------+---------+--------------+-------------+
|Timestamp |Platform |Hashtag       |Region       |
+----------+---------+--------------+-------------+
|2026-03-04|LinkedIn |#Netflix      |Latin America|
|2026-03-04|Facebook |#CloudComputing|North America|
|2026-03-01|Instagram|#Netflix      |Asia-Pacific |
|2026-03-04|Facebook |#CloudComputing|Middle East  |
|2026-03-01|LinkedIn |#Tech         |Middle East  |
|2026-03-01|LinkedIn |#Movies       |Europe       |
|2026-03-01|LinkedIn |#Hollywood    |Europe       |
|2026-03-04|LinkedIn |#Netflix      |North America|
|2026-03-04|LinkedIn |#AI           |North America|
|2026-03-05|Twitter/X|#AI           |North America|
+----------+---------+--------------+-------------+
only showing top 10 rows
```

Step 5: Validating Dataset Structure

```
============================================================
 TASK 4: VALIDATING DATASET STRUCTURE
============================================================
DataFrame Schema:
root
 |-- Timestamp: date (nullable = true)
 |-- Platform: string (nullable = true)
 |-- Hashtag: string (nullable = true)
 |-- Region: string (nullable = true)

Number of Columns : 4
Number of Records : 650
Column Names      : Timestamp, Platform, Hashtag, Region
```

Step 6: Grouping Hashtags

```
============================================================
 TASK 5: GROUPING HASHTAGS
============================================================
Hashtag Occurrences Grouped (Top 15):
+---------------+-----+
|Hashtag        |count|
+---------------+-----+
|#AI            |75   |
|#Cricket       |68   |
|#MachineLearning|58   |
|#Netflix       |52   |
|#DataScience   |45   |
|#Football      |42   |
|#Gaming        |38   |
|#Tech          |35   |
|#IPL           |34   |
|#CloudComputing|32   |
|#CyberSecurity |28   |
|#Movies        |26   |
|#Music         |24   |
|#Sports        |22   |
|#Bollywood     |20   |
+---------------+-----+
only showing top 15 rows
```

Step 7: Counting Hashtag Occurrences

```
================================================================
 TASK 6: COUNTING HASHTAG OCCURRENCES
================================================================
Top 10 Most Frequent Hashtags:
+---------------+-----+
|Hashtag        |count|
+---------------+-----+
|#AI            |75   |
|#Cricket       |68   |
|#MachineLearning|58   |
|#Netflix       |52   |
|#DataScience   |45   |
|#Football      |42   |
|#Gaming        |38   |
|#Tech          |35   |
|#IPL           |34   |
|#CloudComputing |32   |
+---------------+-----+
```

Step 8: Sorting Hashtags by Popularity

```
================================================================
 TASK 7: SORTING HASHTAGS BY POPULARITY
================================================================
Hashtags Sorted by Popularity with Dynamic Rank:
+----+---------------+----------------+
|Rank|Hashtag        |Occurrence_Count|
+----+---------------+----------------+
|1   |#AI            |75              |
|2   |#Cricket       |68              |
|3   |#MachineLearning|58              |
|4   |#Netflix       |52              |
|5   |#DataScience   |45              |
|6   |#Football      |42              |
|7   |#Gaming        |38              |
|8   |#Tech          |35              |
|9   |#IPL           |34              |
|10  |#CloudComputing |32              |
|11  |#CyberSecurity |28              |
|12  |#Movies        |26              |
|13  |#Music         |24              |
|14  |#Sports        |22              |
|15  |#Bollywood     |20              |
+----+---------------+----------------+
only showing top 15 rows
```

Step 9: Identifying Top Trending Hashtags

```
============================================================
 TASK 8: IDENTIFYING TOP TRENDING HASHTAGS
============================================================
Top 5 Trending Hashtags:
+----+----------------+----------------+
|Rank|Hashtag         |Occurrence_Count|
+----+----------------+----------------+
|1   |#AI             |75              |
|2   |#Cricket        |68              |
|3   |#MachineLearning|58              |
|4   |#Netflix        |52              |
|5   |#DataScience    |45              |
+----+----------------+----------------+

[>>>] Top Trending Hashtag: #AI with 75 posts
```

Step 10: Supplementary 1 — Top 3 Trending Hashtags

```
============================================================
 SUPPLEMENTARY 1: TOP 3 TRENDING HASHTAGS
============================================================
+----+----------------+----------------+
|Rank|Hashtag         |Occurrence_Count|
+----+----------------+----------------+
|1   |#AI             |75              |
|2   |#Cricket        |68              |
|3   |#MachineLearning|58              |
+----+----------------+----------------+

Top Trending Hashtag:
#AI (Total Occurrences: 75)
```

Step 11: Supplementary 2 — Daily Trend Comparison

```
================================================================
 SUPPLEMENTARY 2: DAILY TREND COMPARISON
================================================================
Comparing Hashtag Trends Across Multiple Days...

Daily Hashtag Counts (Sample 12 records):
+----------+---------------+-----+
|Timestamp |Hashtag        |count|
+----------+---------------+-----+
|2026-03-01|#AI            |16   |
|2026-03-01|#Netflix       |12   |
|2026-03-01|#Cricket       |11   |
|2026-03-01|#DataScience   |10   |
|2026-03-01|#Movies        |9    |
|2026-03-01|#IPL           |8    |
|2026-03-01|#CloudComputing|7    |
|2026-03-01|#Bollywood     |7    |
|2026-03-01|#Tech          |7    |
|2026-03-01|#Gaming        |6    |
|2026-03-01|#Football      |6    |
|2026-03-01|#MachineLearning|6   |
+----------+---------------+-----+
only showing top 12 rows

Most Popular Hashtag Per Day:
+----------+----------------+-----------+
|Timestamp |Daily_Top_Hashtag|Daily_Count|
+----------+----------------+-----------+
|2026-03-01|#AI             |16         |
|2026-03-02|#AI             |17         |
|2026-03-03|#Cricket        |16         |
|2026-03-04|#AI             |19         |
|2026-03-05|#Cricket        |14         |
+----------+----------------+-----------+
```

Step 12: Supplementary 3 — Regional Hashtag Analysis

```
============================================================
 SUPPLEMENTARY 3: REGIONAL HASHTAG ANALYSIS
============================================================
Total Posts Per Region:
+-------------+-----------+
|Region       |Total_Posts|
+-------------+-----------+
|Asia-Pacific |145        |
|North America|142        |
|Middle East  |122        |
|Europe       |121        |
|Latin America|120        |
+-------------+-----------+

Top Trending Hashtag Per Region:
+-------------+---------------+----------+
|Region       |Top_Hashtag    |Post_Count|
+-------------+---------------+----------+
|Asia-Pacific |#Cricket       |20        |
|Latin America|#MachineLearning|20        |
|North America|#AI            |20        |
|Middle East  |#AI            |16        |
|Europe       |#MachineLearning|15        |
+-------------+---------------+----------+
```

Step 13: Task 10 / Supplementary 4 — User Interest and Hashtag Categories

```
============================================================
 TASK 10 & SUPPLEMENTARY 4: USER INTEREST & HASHTAG CATEGORIES
============================================================
User Interest Distribution by Category:
+-------------+---------------+
|Category     |Number_of_Posts|
+-------------+---------------+
|Technology   |273            |
|Sports       |199            |
|Entertainment|178            |
+-------------+---------------+

[>>>] Most Popular Category: Technology with 273 posts

Platform Distribution:
+---------+-----------+
|Platform |Total_Posts|
+---------+-----------+
|Facebook |172        |
|Instagram|165        |
|LinkedIn |160        |
|Twitter/X|153        |
+---------+-----------+
```

Step 14: Supplementary 5 — Trend Visualization

```
============================================================
 ==== SUPPLEMENTARY 5: TREND VISUALIZATION
============================================================
Creating Matplotlib Bar Chart for Top 10 Trending Hashtags...
Chart successfully saved to: C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-9\hashtag_trends.png

============================================================
 ==== BUSINESS RECOMMENDATIONS
============================================================
[DISCLAIMER]: Educational analysis based on generated/sample social media data.

1. Campaign Targeting : Focus promotional analysis on high-engagement Technology and Sports hashtags.
2. Platform Strategy : Adapt content distribution according to the platform where relevant
technology and sports discussions are concentrated.
3. Emerging Topics : Track daily changes in hashtag frequency to identify rapidly growing discussions.
4. Geo-Targeted Content : Adjust campaign themes according to regional interests, such as cricket discussions
   in Asia-Pacific.
5. Community Engagement : Use interactive content around frequently observed topics such as #AI, #Cricket
   and #Netflix.

============================================================
 ==== FINAL SOCIAL MEDIA TREND ANALYSIS REPORT
============================================================
================================================================================
=====
====
        TRENDPULSE SOCIAL MEDIA TREND DETECTION REPORT
================================================================================
=====
====
1. DATASET DETAILS:
  - Source File          : social_media_hashtags.csv
  - Total Records          650
  - Schema               : Timestamp (string), Platform (string), Hashtag (string), Region (string)
  - Data Nature            : Generated / Sample dataset for educational evaluation

2. TOTAL RECORDS:
  - Total Monitored Posts      650

3. UNIQUE HASHTAGS:
  - Total Distinct Hashtags    18

4. PLATFORM DISTRIBUTION:
Platform       | Total Posts
-----------------------
Facebook       | 172
```

Instagram | 165
LinkedIn | 160

Twitter/X | 153

5. REGION DISTRIBUTION:

Region	Top Hashtag	Post Count
Asia-Pacific	#Cricket	20
Latin America	#MachineLearning	20
North America	#AI	20
Middle East	#AI	16
Europe	#MachineLearning	15

6. HASHTAG FREQUENCY ANALYSIS:

- Hashtag frequency was evaluated across the monitored sample using grouping, counting and descending frequency sorting.

7. TOP 10 HASHTAGS:

Rank	Hashtag	Occurrences
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58
4	#Netflix	52
5	#DataScience	45
6	#Football	42
7	#Gaming	38
8	#Tech	35
9	#IPL	34
10	#CloudComputing	32

8. TOP 5 TRENDING HASHTAGS:

Rank	Hashtag	Occurrences
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58
4	#Netflix	52
5	#DataScience	45

9. TOP 3 TRENDING HASHTAGS:

Rank	Hashtag	Occurrences
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58

* Top Trending Hashtag : #AI (75 occurrences)

10. DAILY TREND ANALYSIS:

Date	Daily Top Hashtag	Post Count
2026-03-01	#AI	16
2026-03-02	#AI	17
2026-03-03	#Cricket	16
2026-03-04	#AI	19
2026-03-05	#Cricket	14

11. REGIONAL ANALYSIS:

Region	Top Hashtag	Post Count
Asia-Pacific	#Cricket	20
Latin America	#MachineLearning	20
North America	#AI	20
Middle East	#AI	16
Europe	#MachineLearning	15

12. CATEGORY ANALYSIS:

Category	Number of Posts
Technology	273
Sports	199
Entertainment	178

* Leading Category : Technology (273 posts)

13. USER INTEREST ANALYSIS:

- User attention in the sample is concentrated mainly around Technology topics such as #AI and #MachineLearning, followed by Sports discussions including #Cricket and #Football, with Entertainment topics such as #Netflix also appearing prominently.

(#Netflix).

14. BUSINESS RECOMMENDATIONS:

* DISCLAIMER: Educational analysis based on generated/sample social media data.
- Use high-frequency topics such as #AI, #Cricket and #MachineLearning as inputs for campaign planning.
- Select suitable platforms for different audience interests, such as professional technology content on LinkedIn and sports discussions on Twitter/X.
- Observe daily changes in hashtag activity so marketing responses can be adjusted when new topics begin to gain attention.

15. CONCLUSION:

PySpark DataFrame transformations, grouping and analytical operations provide a scalable way to process large social-media datasets and identify useful trends.

The results show how frequency, daily, regional and category-based analysis can support structured business interpretation of social media activity.

Key Questions / Analysis

Q1. Why is Apache Spark suitable for social media analytics?

Answer:

Apache Spark is suitable for social media analytics because it can process large amounts of data efficiently using distributed computing. Spark DataFrames provide fast operations such as filtering, grouping, aggregation and sorting, which are useful for analysing large social media datasets.

Real-time analytics can reveal changes in online activity as they happen. This enables organizations to react more quickly to emerging topics, customer interests and market signals.

Q2. What is the significance of hashtag frequency?

Answer:

Hashtag frequency represents how many times a particular hashtag appears in the dataset. A higher frequency indicates that the topic is receiving more attention in the analyzed sample.

Q3. How can trend analysis support digital marketing campaigns?

Answer:

Trend analysis helps marketers identify popular topics and changing user interests. Businesses can use this information to create relevant content, plan campaigns and monitor emerging topics.

Q4. Why do businesses monitor social media trends?

Answer:

Businesses monitor social media trends to understand customer interests, identify emerging topics, observe discussions about products or services and improve their marketing strategies.

Q5. How can real-time analytics improve decision making?

Answer:

Real-time analytics allows organizations to detect changes in trends quickly. This can help businesses respond faster to emerging topics, customer interests and market changes.

Businesses track social media trends to understand audience interests, discover emerging conversations, observe

reactions to products or services and improve marketing decisions.

Trend analysis shows which subjects are gaining attention and how audience interests change. Marketers can use these observations to create timely content, plan campaigns and watch for new trends.

Result

The Social Media Trend Detection practical was successfully implemented using PySpark.

The dataset contained 650 social media records and 18 distinct hashtags.

Hashtag frequency, popularity ranking, top trends, daily trends, regional trends and category-based user interests were successfully analyzed.

The most frequently occurring hashtag in the sample dataset was #AI with 75 occurrences.

The Technology category had the highest number of posts with 273 posts.

A visualization of the top 10 hashtags was also successfully generated using Matplotlib.

Conclusion

The practical demonstrated how **PySpark DataFrame operations and aggregation techniques** can be used for social media trend detection. Hashtags were grouped, counted and ranked according to their frequency.

Daily and regional analysis provided additional information about changing user interests. Category analysis showed the distribution of Technology, Sports and Entertainment topics.

The practical demonstrates that PySpark can be used to process and analyze large-scale social media datasets efficiently and extract useful trends for business analysis.

Hashtag frequency is the number of times a particular hashtag occurs in the analyzed dataset. A larger count indicates greater attention to that topic within the sample.

Apache Spark is useful for social media analytics because distributed processing can handle large datasets efficiently. Spark DataFrames support filtering, grouping, aggregation and sorting operations needed for trend analysis.